

8(a)(i)	Simple harmonic motion is an oscillatory motion where the acceleration of the object is directly proportional to its displacement from the equilibrium position and directed towards the equilibrium position.	[2]
8(a)(ii)1.	The acceleration is <u>always in opposite direction to the displacement</u> . This indicates the presence of a <u>restoring force/acceleration that pulls the object back to the equilibrium position</u> when it is on either side of the equilibrium position in order for the oscillatory motion to continue.	[1] [1] for correct explanation
8(a)(ii)2.	The line is <u>not straight</u> , indicating a non-linear relationship between acceleration and displacement. The <u>acceleration</u> , (hence the <u>restoring force</u>), is <u>not directly proportional to the displacement</u> which is a condition for simple harmonic motion.	[1] [1] for correct explanation
8(b)(i)1.	A longitudinal wave is one in which the <u>particles/molecules of the medium oscillate parallel</u> to the <u>direction of travel of the wave or direction of propagation of wave energy</u> .	[2]
8(b)(i)2.	The speed of the wave is the distance moved by the waveform per unit time in the direction of travel of the wave.	[1]

No.	Solution	Remarks
8(b)(ii)1.	$T = \frac{1}{f}$ $= \frac{1}{820}$ $= 1.22 \times 10^{-3} \text{ s}$	[1] for correct substitution [1] answer
8(b)(ii)2.	The time interval is half the period $= \frac{1.22 \times 10^{-3}}{2}$ $= 6.1 \times 10^{-4} \text{ s}$	[1]
8(b)(ii)3.	Maximum speed, $v_{\max}$ $= \omega x_0$ $= 2\pi f x_0$ $= 2\pi \times 820 \times 610 \times 10^{-9}$ $= 3.143 \times 10^{-3}$ $\approx 3.14 \times 10^{-3} \text{ m s}^{-1}$	[1] for correct expression and substitution [1] for correct answer
8(b)(ii)4.	Maximum vibrational kinetic energy, $E_{\max}$ $= \frac{1}{2} m v_{\max}^2$ $= \frac{1}{2} 5.3 \times 10^{-26} \quad 3.143 \times 10^{-3}^2$ $= 2.617 \times 10^{-31}$ $\approx 2.62 \times 10^{-31} \text{ J}$	[1] for correct expression and substitution [1] for correct answer
8(b)(iii)	The speed of travel of sound in a gas at room temperature is about 330 m s^{-1} , more than 100,000 times the maximum speed of $3.14 \times 10^{-3} \text{ m s}^{-1}$ for the vibration of the gas molecules. The speed of sound refers to how fast the wave energy (or disturbance) propagates through the gas, not how fast individual molecules oscillate about their equilibrium position.	[1] [1] for explanation
8(c)(i)	The source is the <u>mechanical energy of a vibrating object in contact with the gas</u> , exerting a <u>force parallel to the direction of propagation</u> of the sound wave.	[1]
8(c)(ii)	The restoring force on a molecule in a sound wave arises from <u>pressure differences</u> in the surrounding gas, <u>caused by compressions and rarefactions in the wave</u> .	[1]